Storage and Bandwidth Estimation

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solve this question with proper details

Below is a worked-through example of how one might arrive at ballpark figures for **storage**, **caching**, and **bandwidth** based on the scenario's numbers and a few reasonable assumptions. The key steps are to:

1. **Determine daily tweet volumes (text vs. media).**
2. **Estimate per-tweet sizes** for text and media.
3. **Roll up to total storage over the five-year retention.**
4. **Incorporate the viewer-to-uploader ratio** to estimate bandwidth.
5. **Account for extended caching** (e.g., verified accounts).

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1. Baseline Usage Figures

- **Monthly active users (MAU):** 300 million
- **Daily active users (DAU):** 50% of MAU

$$300\,\text{million} \times 50\% = 150\,\text{million (DAU)}$$

- **Tweets per day (total):** 2 tweets per daily active user

$$150\,\text{million} \times 2 = 300\,\text{million tweets/day}$$

- **Media-bearing tweets:** 10% of all tweets

$$10\% \times 300\,\text{million} = 30\,\text{million media tweets/day}$$

- **Text-only tweets:** 90% of all tweets

$$90\% \times 300\,\text{million} = 270\,\text{million text tweets/day}$$

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2. Storage Estimates

2.1 Text Storage

1. **Tweet size (text only):** up to 240 characters
   - A simplistic assumption is **1 byte per character** (ignoring metadata).
   - Hence each text tweet ≈ 240 bytes.
2. **Total text volume per day:**

$$270 \text{ million tweets/day} \times 240 \text{ bytes/tweet} = 64.8 \times 10^9 \text{ bytes/day} \approx 64.8 \text{ GB/day}.$$

3. **Five-year retention (text):**

$$64.8 \text{ GB/day} \times 365 \text{ days/year} \times 5 \text{ years} \approx 118,260 \text{ GB} \approx 118 \text{ TB (over 5 years)}.$$

> **Note:** Real systems store extra metadata, indexes, and replicas, so the *actual* text footprint is often several times this raw figure.

2.2 Media Storage

1. **Media tweets per day:** 30 million.
2. **Assumed average media size:** 2 MB per photo/video.

- In practice, videos vary widely, but 2 MB is a reasonable *average* guess for short clips/images.

3. **Total media volume per day:**

$$30 \text{ million media tweets/day} \times 2 \text{ MB/tweet} = 60 \text{ million MB/day} = 60,000 \text{ GB/day} = 60 \text{ TB/day}.$$

4. **Five-year retention (media):**

$$60 \text{ TB/day} \times 365 \times 5 = 60 \times 1825 \text{ TB} = 109,500 \text{ TB} = 109.5 \text{ PB (over 5 years)}.$$

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3. Caching Requirements

- The prompt says: “Posts from verified accounts are cached for an extended period of one year.”
- The fraction of tweets from verified accounts is *not* specified, but if you assume (for example) **1%** of tweets come from verified users, then:

- **Verified tweets per day:**

$$1\% \times 300\,\text{million} = 3\,\text{million/day}.$$

- Of these, **10%** contain media → 300 k media tweets/day; **90%** are text → 2.7 million text tweets/day.
- **Text volume (verified/day):**

$$2.7\,\text{million} \times 240\,\text{bytes} \approx 648\,\text{MB/day}.$$

- **Media volume (verified/day):**

$$300,000 \times 2\,\text{MB} = 600\,\text{MB/day}.$$

- **Total verified data/day:** $\approx 1.25\,\text{GB/day}.$

  Over one year, that is $\sim 450\,\text{GB}$ if only 1% are verified. If more are verified (say 5%), simply multiply accordingly.

- In practice, you would size your high-speed cache (e.g., memory or specialized CDN nodes) to hold a rolling window of verified content for that one-year period, plus overhead for hot content.

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4. Bandwidth Requirements

The scenario notes a **viewer-to-uploader ratio of 3:1**, meaning that for each piece of content uploaded, on average there are 3 downloads/views.

4.1 Text Tweets

- **Uploads (text/day):** ~64.8 GB (as calculated above).
- **Downloads (3× the upload volume):** ~194.4 GB/day.
- **Total text data transferred/day:**

$$64.8\,\text{GB (upload)} + 194.4\,\text{GB (download)} \approx 259.2\,\text{GB/day}.$$

4.2 Media Tweets

- **Uploads (media/day):** ~60 TB (as calculated above).
- **Downloads (3× the upload volume):** ~180 TB/day.
- **Total media data transferred/day:**

$$60\,\text{TB (upload)} + 180\,\text{TB (download)} = 240\,\text{TB/day}.$$

4.3 Overall Daily Bandwidth

Summing text + media:

$$\underbrace{240\,\mathrm{TB/day}}_{\text{media}} + \underbrace{0.26\,\mathrm{TB/day}}_{\text{text}} \approx 240.26\,\mathrm{TB/day}.$$

Most of the bandwidth is dominated by media traffic. Over a month (30 days), that is $\approx 7.2\,\mathrm{PB/month}$ of data transfer.

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5. Putting It All Together

1. **Storage (5-year retention):**
   - Text: ~118 TB total over 5 years
   - Media: ~109.5 PB total over 5 years
2. **Cache:**
   - Verified tweets (example 1% assumption): ~1.25 GB/day of "verified" data; $\approx 450\,\mathrm{GB/year}$.
   - Would need a cache sized to hold up to a year's worth of hot verified content plus overhead.
3. **Bandwidth:**
   - **Daily:** ~240 TB/day total (dominated by media).
   - **Monthly:** ~7.2 PB/month.
4. **Query Rate:**
   - The system handles ~3500 queries/second (QPS).
   - Actual read/write distribution and concurrency would drive how you provision front-end servers, load balancers, and database nodes.

Final Observations

- **Media** dwarfs **text** both in storage and bandwidth requirements.
- **Caching** is crucial for performance, especially for verified or high-traffic accounts, to avoid the slower 10 ms network-side fetches whenever possible (as opposed to ~250 µs from memory).
- Real-world deployments would incorporate replication, backups, indexing overhead, and compression—so actual provisioning often exceeds these raw estimates by a factor.

These figures, while approximate, provide a reasonable **order-of-magnitude** foundation for designing "Chirp's" infrastructure.